



GLOBAL COFFEE HEALTH ADVANCED VISUALIZATION

Capstone #2

M2M Tech Connect
Data Talent Program
Cohort 18

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OBJECTIVE

- Visualize the data via charts using Python and Bokeh
- Visualize the data in a world map using Geo JSON and Bokeh
- Export the visualization to Netlify
- Tools used:
 - Google Colab Python Notebook
 - Pandas – data manipulation
 - Bokeh – data visualization

DATASET OVERVIEW

Source:

<https://www.kaggle.com/datasets/uom190346a/global-coffee-health-dataset>

Num. of rows: 10,000

Num. of columns: 15

Notes:

- All columns are filled in except for “Health Issues” which have null values

Index: 10000 entries, 1 to 10000

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	Age	10000 non-null	int64
1	Gender	10000 non-null	object
2	Country	10000 non-null	object
3	Coffee_Intake	10000 non-null	float64
4	Caffeine_mg	10000 non-null	float64
5	Sleep_Hours	10000 non-null	float64
6	Sleep_Quality	10000 non-null	object
7	BMI	10000 non-null	float64
8	Heart_Rate	10000 non-null	int64
9	Stress_Level	10000 non-null	object
10	Physical_Activity_Hours	10000 non-null	float64
11	Health_Issues	4059 non-null	object
12	Occupation	10000 non-null	object
13	Smoking	10000 non-null	int64
14	Alcohol_Consumption	10000 non-null	int64

dtypes: float64(5), int64(4), object(6)

memory usage: 1.2+ MB

DATASET OVERVIEW

Column	Type	Description
ID	Integer	Unique record ID (1–10000)
Age	Integer	Age of participant (18–80 years)
Gender	Categorical	Male, Female, Other
Country	Categorical	Country of residence (20 countries)
Coffee_Intake	Float	Daily coffee consumption in cups (0–10)
Caffeine_mg	Float	Estimated daily caffeine intake in mg (1 cup ≈ 95 mg)
Sleep_Hours	Float	Average hours of sleep per night (3–10 hours)
Sleep_Quality	Categorical	Poor, Fair, Good, Excellent (based on sleep hours)

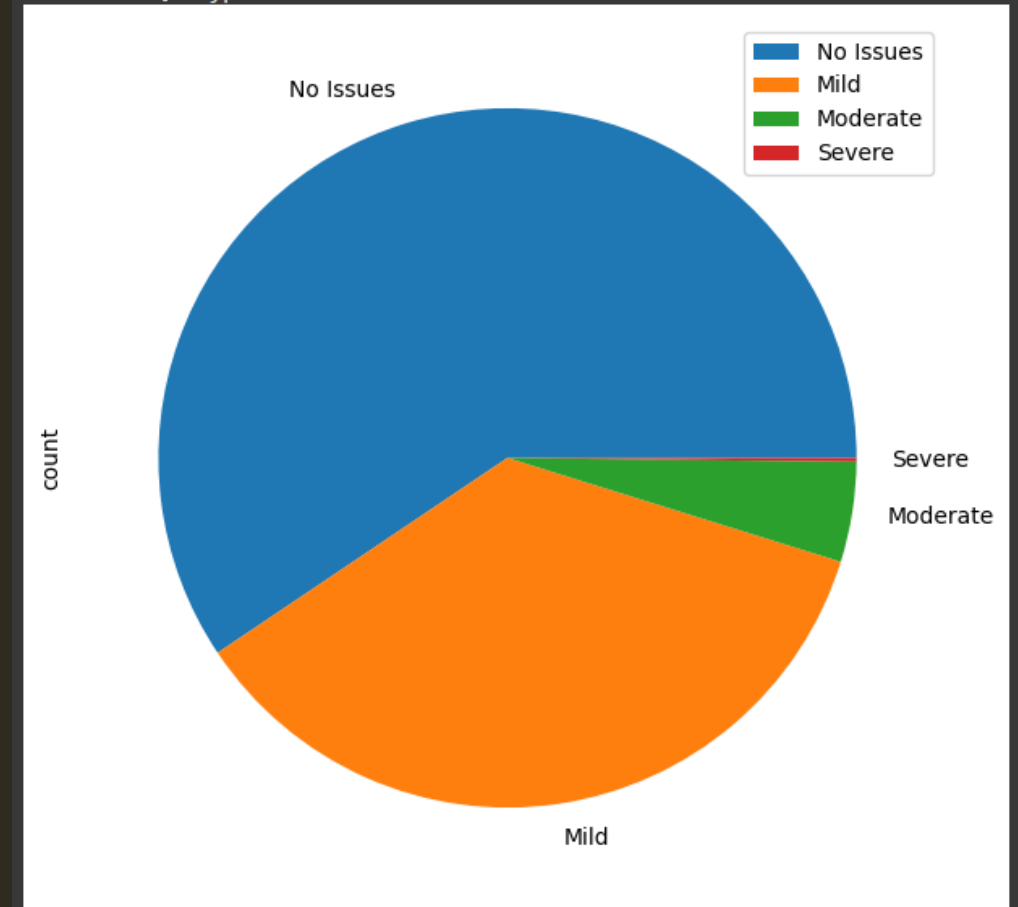
Column	Type	Description
BMI	Float	Body Mass Index (15–40)
Heart_Rate	Integer	Resting heart rate (50–110 bpm)
Stress_Level	Categorical	Low, Medium, High (based on sleep hours and lifestyle)
Physical_Activity_Hours	Float	Weekly physical activity (0–15 hours)
Health_Issues	Categorical	None, Mild, Moderate, Severe (based on age, BMI, and sleep)
Occupation	Categorical	Office, Healthcare, Student, Service, Other
Smoking	Boolean	0 = No, 1 = Yes
Alcohol_Consumption	Boolean	0 = No, 1 = Yes

DATA VISUALIZATION

AGE GROUP AND HEALTH ISSUES VS CAFFEINE INTAKE (MG)

- Get the Caffeine Intake (mg) consumed by each individual based on their Health Issue category and Age Group
- The data will be grouped by Age Group, and each Age Group will contain each unique Health Issue category (No Issues, Mild, Moderate, Severe)
- The plot pie on the right shows an overview of the count of each Health Issue category in our data set

```
Health_Issues
No Issues    5941
Mild         3579
Moderate     463
Severe       17
Name: count, dtype: int64
```



DATA VISUALIZATION

AGE GROUP AND HEALTH ISSUES VS CAFFEINE INTAKE (MG)

➤ The Age Group will be determined by decade as shown in the table on the right

Age Group	Ages Covered
10s (teens)	10 – 19 yrs. old
20s	20 – 29 yrs. old
30s	30 – 39 yrs. old
40s	40 – 49 yrs. old
50s	50 – 59 yrs. old
60s	60 – 69 yrs. old
70s	70 – 79 yrs. old
80s	80 – 89 yrs. old

FORMING THE AGE GROUP

Create a function that will return the Age Group based on the Age

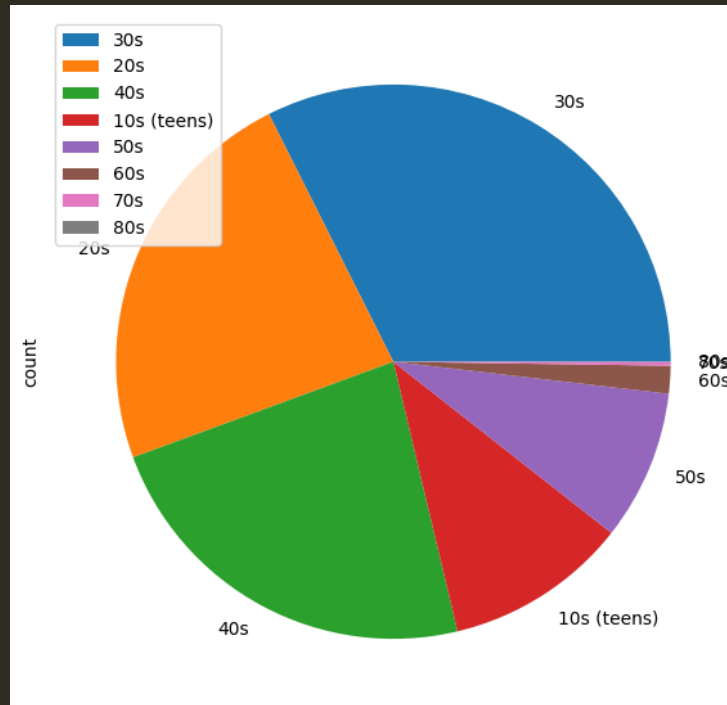


Apply the function to the Age column in the data frame



Assign the result to a new data frame column named Age Group

Age Group	Ages Covered
10s (teens)	10 – 19 yrs. old
20s	20 – 29 yrs. old
30s	30 – 39 yrs. old
40s	40 – 49 yrs. old
50s	50 – 59 yrs. old
60s	60 – 69 yrs. old
70s	70 – 79 yrs. old
80s	80 – 89 yrs. old

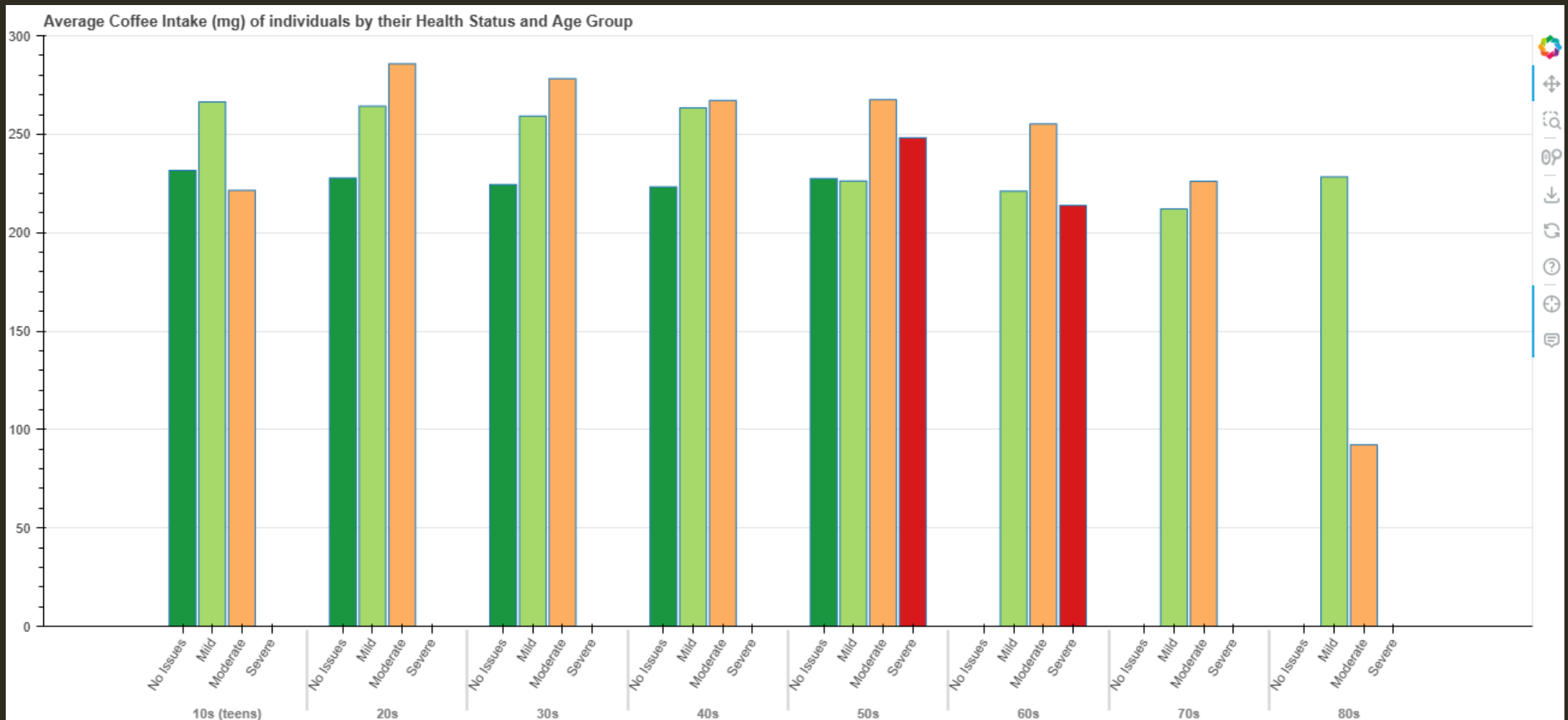


```
1 age_group_count = df_coffee['Age_Group'].value_counts()  
2 age_group_count.plot.pie(legend=True, figsize=(7,7))  
3 print(age_group_count)
```

```
Age_Group  
30s      3241  
20s      2320  
40s      2311  
10s (teens) 1068  
50s       875  
60s       162  
70s        21  
80s         2  
Name: count, dtype: int64
```

DATA VISUALIZATION

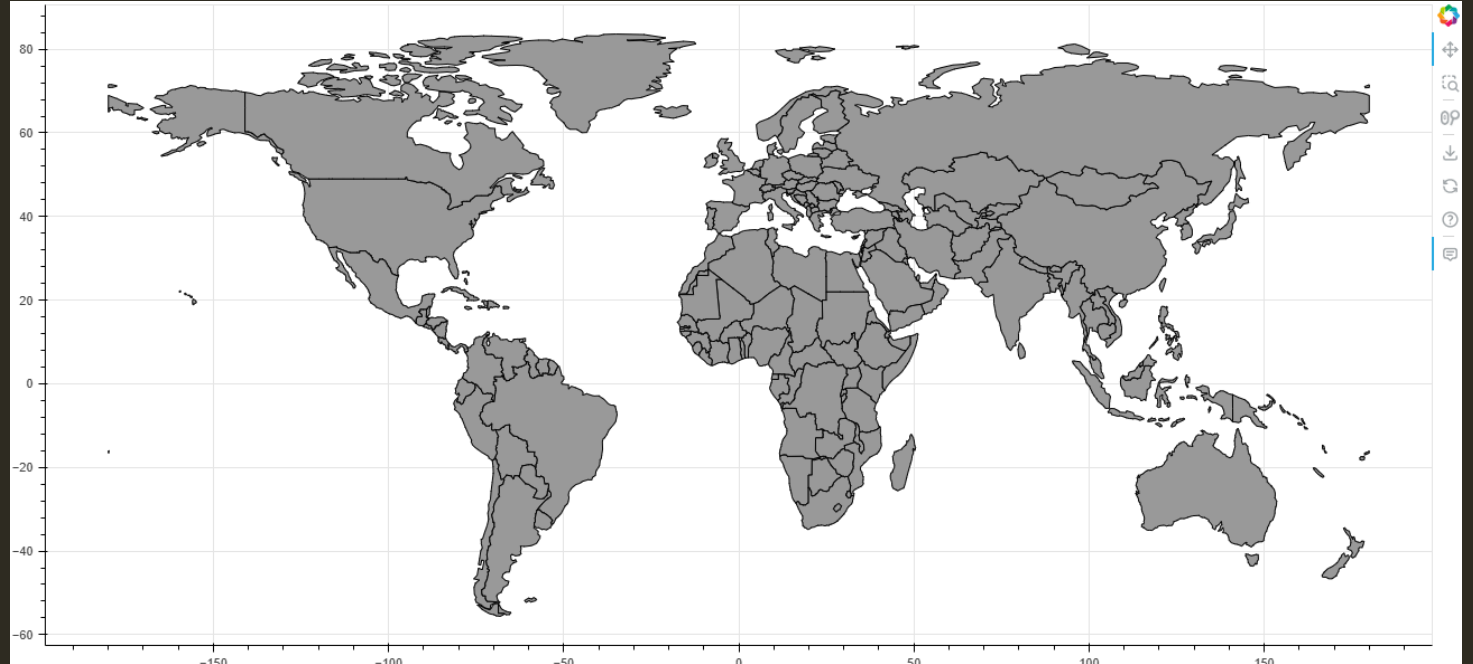
AGE GROUP AND HEALTH ISSUES VS CAFFEINE INTAKE (MG)



DATA VISUALIZATION

AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY

- Get the average Caffeine Intake (mg) consumed by each Country and visualize using a world map
- The world map will be shown using Geo JSON
- We will get the world map Geo JSON file from the following source:
<https://geojson-maps.kyd.au/>
- Shown on the right is how the world map will look like after importing the Geo JSON file and showing it to a Bokeh plot
- Our goal is to change the color of the countries that has existing data and add a hover tool for more details



INVOLVED COUNTRIES

- When we group our data frame by Country, we will see how many countries we have data on
- After calling SIZE on the group object, we will see that we have 20 countries
- We should expect a lot of grey areas in our visualization since we are only highlighting 20 countries
- Note that UK and USA spellings are abbreviated

```
1 world_country_group = df_coffee.groupby('Country')
2 print('Number of groups', world_country_group.ngroups, '\n')
3 print(world_country_group.size())
```

Number of groups 20

Country	
Australia	497
Belgium	497
Brazil	456
Canada	543
China	521
Finland	510
France	499
Germany	497
India	524
Italy	509
Japan	469
Mexico	483
Netherlands	494
Norway	523
South Korea	512
Spain	486
Sweden	513
Switzerland	500
UK	519
USA	448

dtype: int64

EXPLORING THE WORLD MAP'S JSON FILE

- The world map's JSON file that we imported is a type of 'Feature Collection' and contains the following layers:
 - ['features'] will be a list of features, with a total of 175 items representing each country
 - Each feature will contain 'type', 'properties' and 'geometry' objects
 - ['properties'] is a very long dictionary containing the properties of the feature. We will be needing the 'name' property to match our data. We will also modify the dictionary to add more properties such as Color, Rank and the Average Caffeine Intake
 - ['geometry'] is a Polygon type which Bokeh uses to draw our world map plot, we won't be touching this
- Since we know how the JSON file looks like, we will access the properties in Python like this:
 - Geojson['features'][i]['properties']
 - 'i' will be the index in the 'features' list

```
type: "FeatureCollection"
features:
  0:
    type: "Feature"
    properties: { featurecla: "Admin-0 country", scalerank: 1, labelrank: 5, _ }
    geometry:
      type: "Polygon"
      coordinates: [ (36)[...] ]
    1: { type: "Feature", properties: {...}, geometry: {...} }
    2: { type: "Feature", properties: {...}, geometry: {...} }
    3: { type: "Feature", properties: {...}, geometry: {...} }
    4: { type: "Feature", properties: {...}, geometry: {...} }
    labelrank: 5
    sovereignt: "Nicaragua"
    sov_a3: "NIC"
    adm0_dif: 0
    level: 2
    type: "Sovereign country"
    tlc: "1"
    admin: "Nicaragua"
    adm0_a3: "NIC"
    geou_dif: 0
    geounit: "Nicaragua"
    gu_a3: "NIC"
    su_dif: 0
    subunit: "Nicaragua"
    su_a3: "NIC"
    brk_diff: 0
    name: "Nicaragua"
    name_long: "Nicaragua"
```

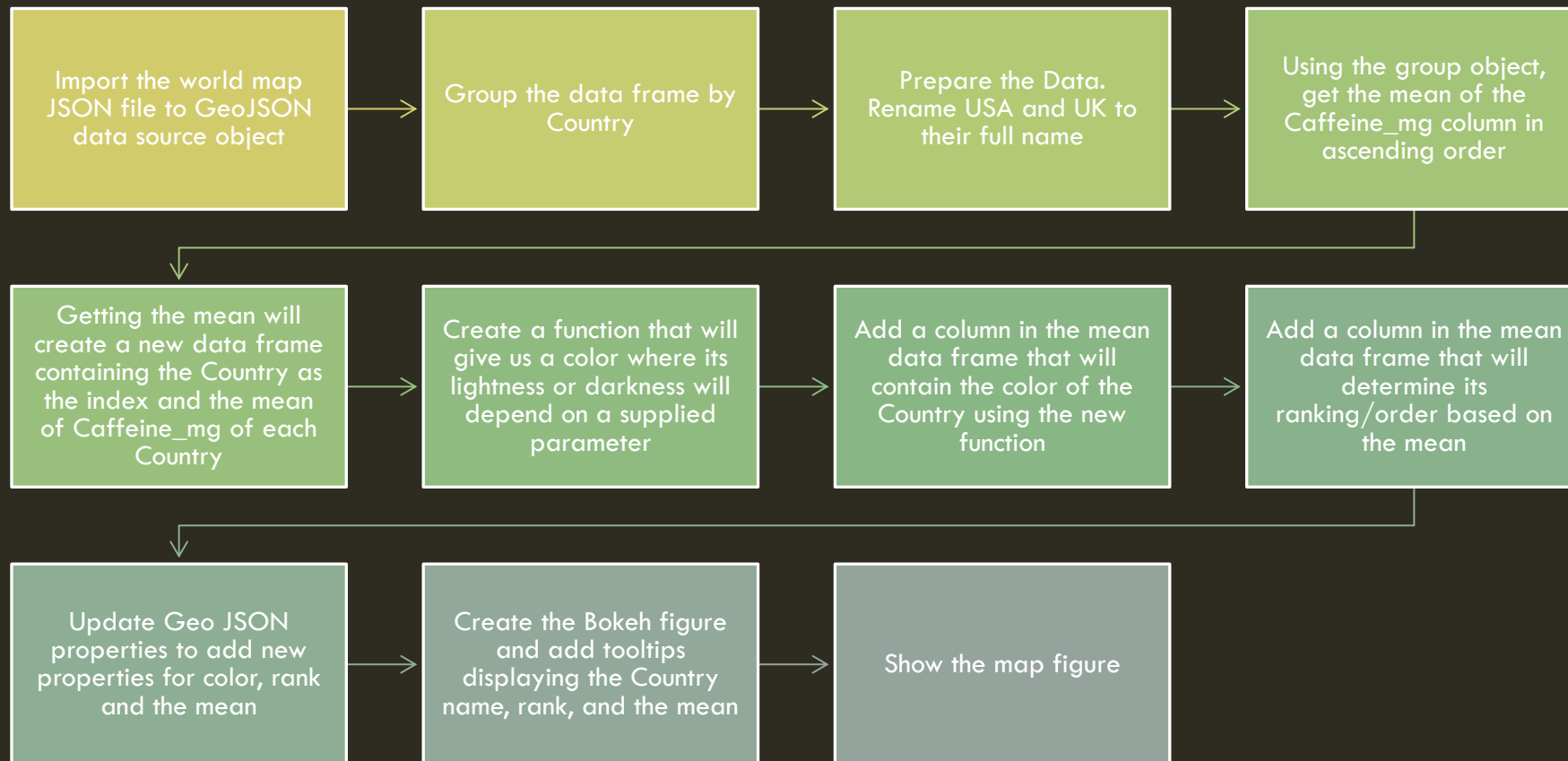
DETERMINE THE COLOR BASED ON RANKING

- To color our world map, we will be using Yellow as a theme
- Our goal is to change the color's darkness based on the Country's average Caffeine intake
- We will accomplish this by making a new function where it will take an amount parameter that will determine the color's darkness
- The Python library used is called 'seaborn' and the figure is called 'palplot'



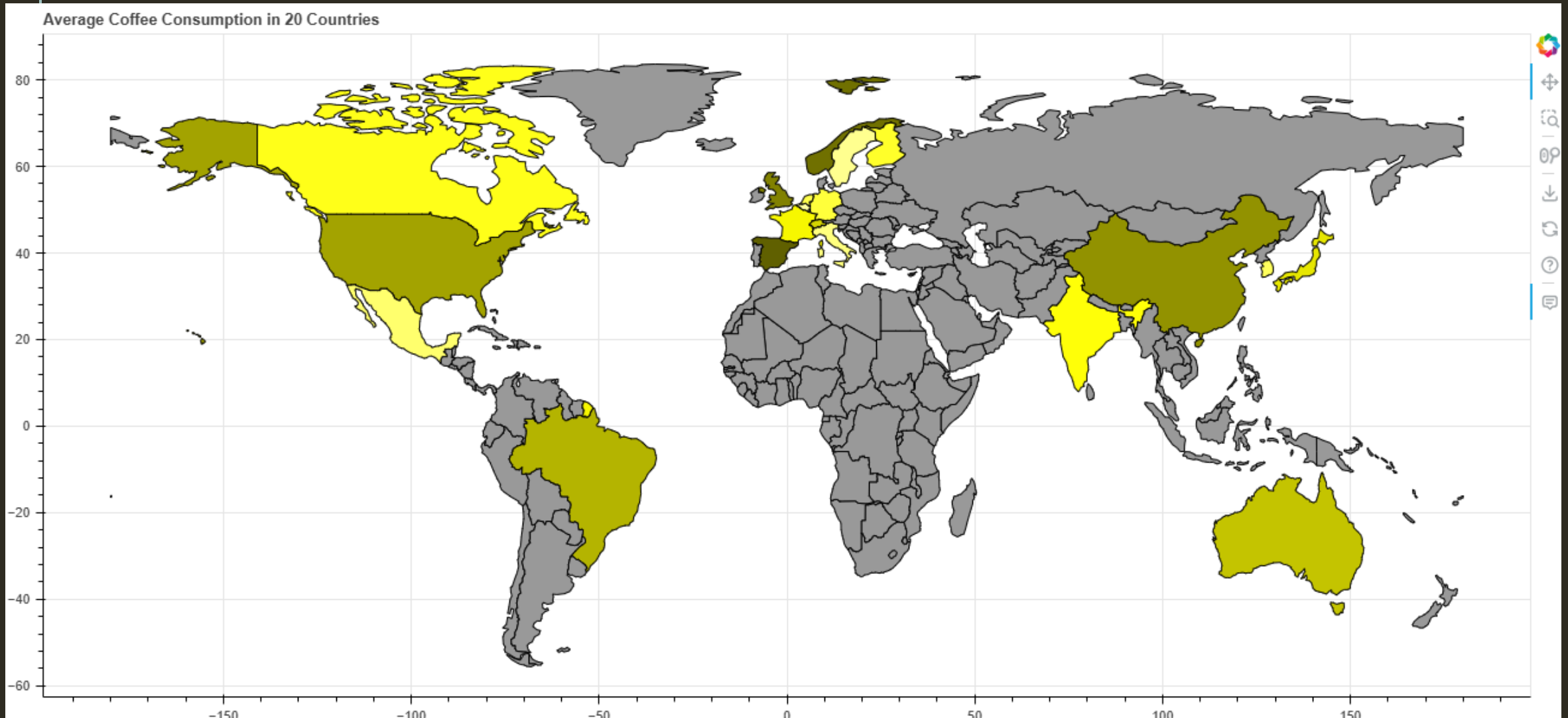
AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY

CODE PROCESS



DATA VISUALIZATION

AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY



PREPARE TO PUBLISH

- In total we have shown 5 Bokeh figures
- To show all these figures into one, we will use Bokeh's layout functionalities:
https://docs.bokeh.org/en/latest/docs/user_guide/basic/layouts.html#general-grid-layout
- Then we will use Bokeh's output file to export the figures in an HTML file
- The platform that we will be using to publish our web page is Netlify:
<https://aramirez-capstone1-0978da.netlify.app/>

```
1 from bokeh.layouts import layout
2 from bokeh.plotting import output_file
3
4 final_layout = layout([
5     [p_age_gender_size, p_age_gender_avgCaf],
6     [p_age_avgCaf],
7     [p_age_health_avgCaf],
8     [p_map]
9 ])
10
11 output_file('index.html')
12
13 show(final_layout)
```

ANALYSIS

➤ Age Group and Health Issue Category vs Caffeine Intake (mg) Comparison

- The 'Severe' Health Issue Category does not have enough sample size, 11 in the 50s age group and 6 in the 60s age group
- From Age Groups 20s to 40s, those who consume more than 250 mg have Mild to Moderate Health Issues
- Those who does not have Health Issues consumes around 200 mg to 250 mg
- The difference in consumption of those with no health issues to those that have is only around 50 mg, which suggests that Coffee consumption might have less impact on one's health, however, consuming less than 250 mg would be a safe measure

➤ Average Coffee Consumption by Country

- Spain ranks number 1 with an average of 247 mg per day
- Belgium ranks last at number 20 with an average of 230 mg per day

SOURCES

Dataset

<https://www.kaggle.com/datasets/uom190346a/global-coffee-health-dataset>

Geo JSON of World Map

<https://geojson-maps.kyd.au/>

Coding Help

Lighten Color - <https://stackoverflow.com/questions/37765197/darken-or-lighten-a-color-in-matplotlib>

Bokeh Grid Layout -

https://docs.bokeh.org/en/latest/docs/user_guide/basic/layouts.html#general-grid-layout



A top-down photograph of a cup of coffee with latte art, served on a light blue saucer and a matching cup. The coffee is on a dark wooden table. A silver spoon is placed to the left of the cup. The text "THANK YOU" is overlaid in white, bold, sans-serif font across the center of the image. A thin white vertical line is positioned to the right of the text.

THANK YOU